# Ecommerce Sales Analysis

Uncovering spending patterns, sales behavior, and customer insights from **1,480 transactions** to guide strategic business decisions.

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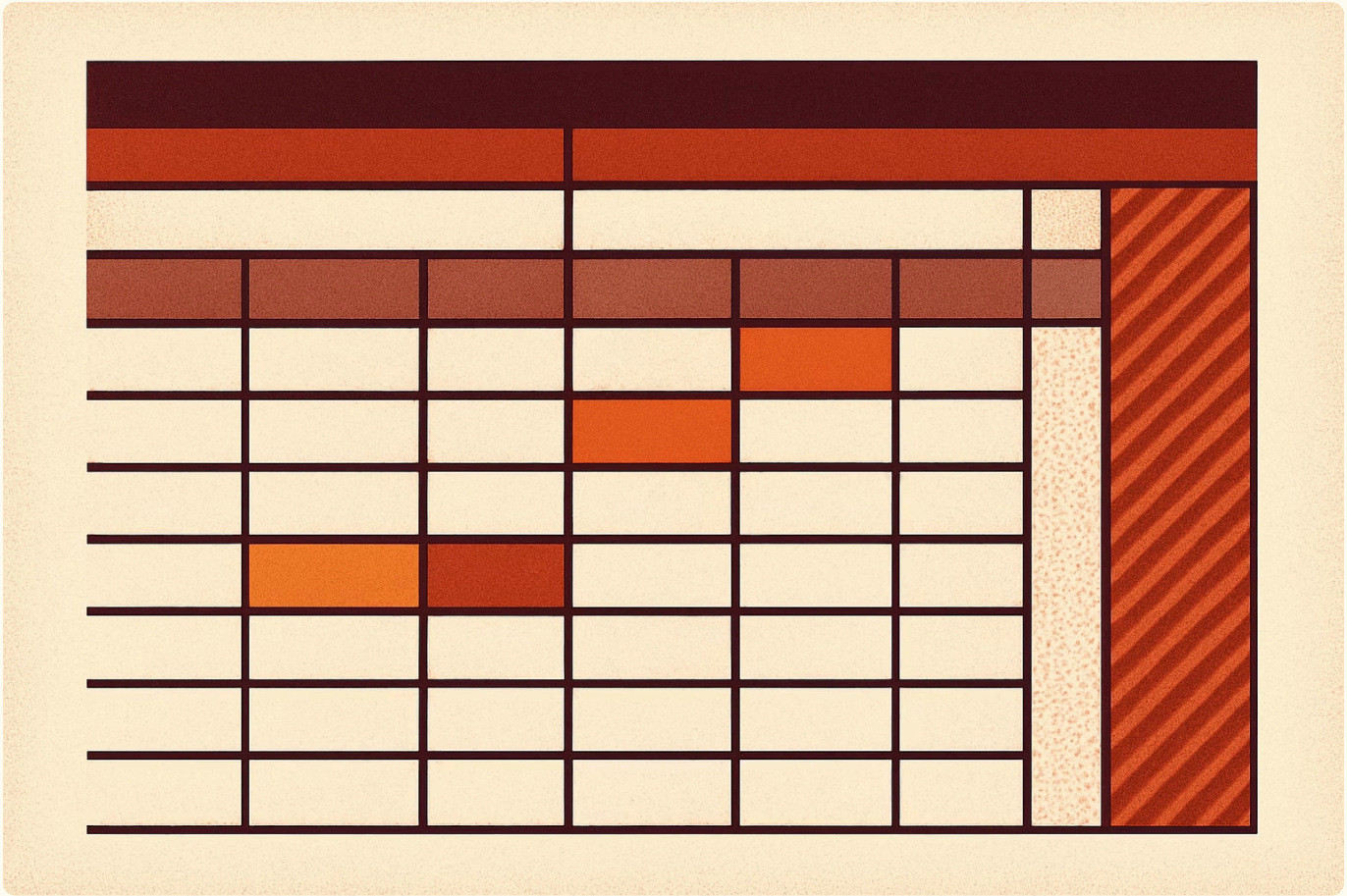
# Dataset at a Glance



## 1,480 Rows · 10 Columns

Transactional records covering demographics, sales details, and business metrics.

## Key Features

Order ID, product name/category, payment method, region, customer satisfaction & rating.

## Missing Data

68 missing values in Customer Rating; 59 missing in Price column.

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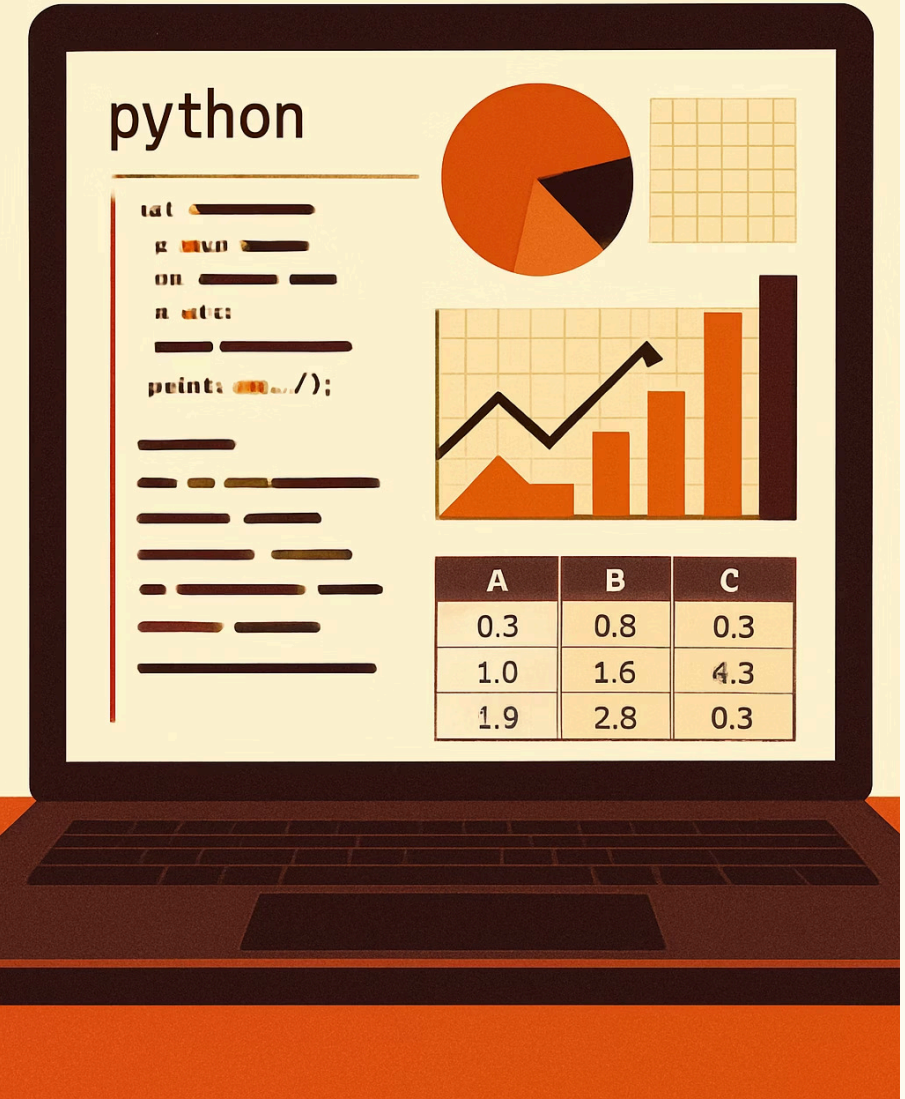


# Exploratory Data Analysis in Python

Data preparation and cleaning performed using **Python IDLE with pandas**.

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# Data Cleaning Steps

## Data Loading

Imported dataset using pandas; explored structure with `df.info()` and `.describe()`.

## Missing Values

Imputed missing Price and Customer Rating values using **median per product category**.

## Standardization

Renamed columns to snake_case; standardized rows to Proper Case; fixed float/string issues in Quantity.

## Database Integration

Connected Python to **MySQL** and loaded the cleaned DataFrame for SQL analysis.

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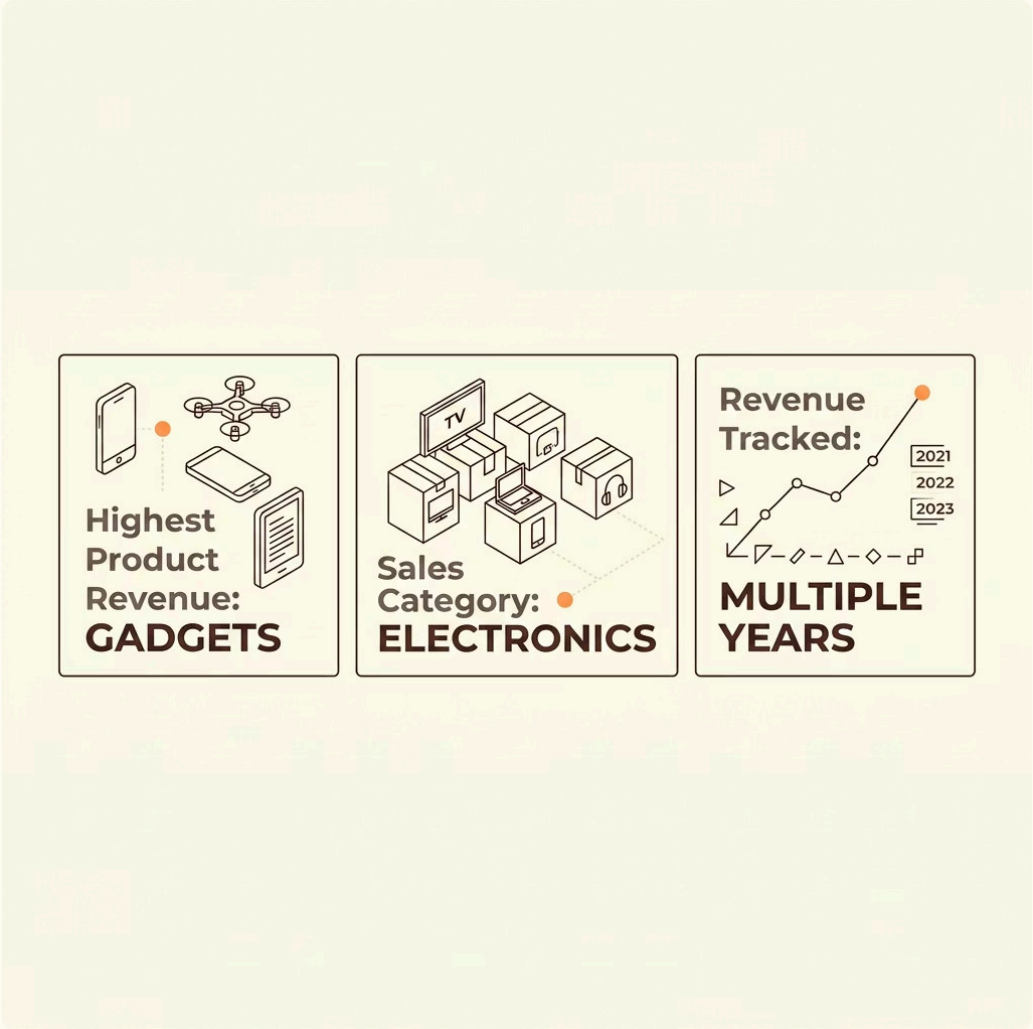
# SQL Analysis: Revenue & Sales




## Key Revenue Queries

- **Highest Revenue Product** — compared total purchase by product name
- **Top Sales Category** — identified product category vs. revenue
- **Revenue by Year** — tracked total revenues across years

ⓘ All queries executed in MySQL after loading the cleaned Python DataFrame.

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# SQL Analysis: Region & Orders

## Regional & Order Insights

- **Highest Sales by Region** — identified top-performing region by sales rate
- **Average Order Value** — compared all transactions by average order
- **Region-wise Order Count** — calculated number of orders per region

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# SQL Analysis: Payments & Ratings

### Most Used Payment Method

Identified which payment method customers use most frequently across all transactions.

### Payment vs. Avg. Cost

Classified sales into revenue segments based on payment method to find highest average cost.

### Customer Rating Range

Identified minimum and maximum customer ratings per transaction.

### Rating by Category

Compared average customer rating across product categories to find highest-rated.

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# Power BI Dashboard

An **interactive dashboard** was built in Power BI to present all key insights visually — enabling dynamic exploration of sales, regions, payments, and customer ratings.

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# Business Recommendations

## Boost Gadgets

Promote exclusive products in the **Gadgets** category — the top revenue driver.

## Loyalty via UPI

Reward **UPI payment** users with loyalty perks to drive higher sales volume.

## Discount Policy

Balance sales boosts with **margin control** through a reviewed discount strategy.

## Targeted Marketing

Focus on **high-revenue segments** and highlight top-rated, best-selling products in campaigns.

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# Key Takeaways

## 1,480 Transactions Analyzed

Cleaned, standardized, and loaded into MySQL via Python.

## 10 SQL Business Queries

Covering revenue, regions, payments, orders, and customer ratings.

## Actionable Insights

Gadgets lead revenue; UPI dominates payments; targeted marketing can maximize ROI.

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